

CLASS 9 MATHEMATICS

Complete Geometry Chapter Study Session & Master Question Bank

Designed for Academic Excellence, Conceptual Rigour, and CBSE Board Answer-Writing Formats

1. The CBSE Answer-Writing Protocol: How to Structure Proofs

Writing a geometric proof is not just about arriving at the correct final step; it is about building a watertight logical argument. To score 100% marks in CBSE board exams, every single step in your proof must follow a standardized structure. Examiners look for specific heading blocks.

Perfect Answer-Writing Framework (The 5-Step Block)

A high-scoring geometry proof must always consist of the following five distinct components:

- Step 1: The Diagram:** Always draw a clean, neat figure using a sharp pencil and ruler on the right-hand side of your answer page. Label all vertices, center (O), chords, and angles clearly.
- Step 2: Given:** Explicitly list the known parameters and conditions. Use mathematical symbols (e.g., write “Given: $AB = CD$ ” instead of “it is given that the two chords are equal”).
- Step 3: To Prove / To Find:** Clearly state what needs to be established mathematically (e.g., “To Prove: $\angle AOB = \angle COD$ ”).
- Step 4: Construction (Optional):** If your proof requires drawing extra lines, dropping perpendiculars, or extending segments, write it down under this block (e.g., “Construction: Join center O to endpoints A and B ”).
- Step 5: Proof (Logical Argumentation):** The core body of your answer. Present it as a systematic series of statements on the left, with the exact mathematical justifications or theorems written in brackets on the right.

Practical Example: How to Write the Justifications

When proving triangles congruent to establish a circle theorem:

- **Incorrect:** $OA = OB$ because they are equal lines.
- **Correct:** $OA = OB$ ($\because$ Radii of the same circle, which are always equal)
- **Incorrect:** $\angle AMC = \angle BMC = 90^\circ$ by common sense.
- **Correct:** $\angle AMC = \angle BMC = 90^\circ$ ($\because$ Linear Pair Axiom, and $\angle AMC = \angle BMC$ established via CPCT)

2. Mathematical Signs and Symbols Glossary

In geometry, writing full sentences for every logical link is time-consuming and cluttering. Instead, we use universally recognized mathematical symbols. Below is the master glossary of symbols required for Class 9 Circle proofs, along with their precise exam usage rules.

Symbols Glossary for CBSE Geometry Proofs

Symbol	Meaning	When/Why to Use	Exam Example
$\therefore$	Therefore	Used to state a logical consequence or the final deduction of a step.	Since $\triangle OAM \cong \triangle OBM$, $\therefore AM = BM$ (by CPCT).
$\because$	Since / Because	Used to introduce a starting premise, given fact, or a theorem as a reason.	$OA = OB$ ($\because$ Radii of the same circle).
$\cong$	Is Congruent To	Used to show that two triangles are completely identical in shape and size.	$\triangle OMA \cong \triangle OMB$ (by RHS congruence rule).
$\perp$	Is Perpendicular To	Used to show that two lines meet at a perfect right angle (90°).	$OM \perp AB$ ($\perp$ Perpendicular from center).
$\parallel$	Is Parallel To	Used to show that two chords or lines lie on the same plane and never intersect.	$AB \parallel CD$ (Given).
$\angle$	Angle	Used as shorthand prefix before naming a geometric angle.	$\angle AOB = 60^\circ$.
$\triangle$	Triangle	Used as shorthand prefix before naming a triangle.	In right-angled $\triangle OMC$:
$\sim$	Is Similar To	Used to show that two shapes have the same shape but different sizes (concentric circles, etc.).	$\triangle ABC \sim \triangle ADE$.
$\widehat{AB}$	Arc AB	Represents a curved portion of the circle's boundary.	If $\widehat{AB} \cong \widehat{CD}$, then chord $AB =$

The Golden Rule of Justification

Never write a mathematical statement in your proof without appending a bracket starting with the “since” (...) sign.

Example:

$AM = \frac{1}{2}AB$ ($\because$ The perpendicular from the center of a circle to a chord bisects the chord).

3. The Circles Master Question Bank

This master question bank contains exactly **45 graded questions** covering the complete syllabus of Class 9 Circles (including concepts from both NCERT and CBSE Exemplar resources). It is structured into four progressive categories to build confidence and ultimate mastery.

Part A: Conceptual Recall & True/False with Justifications (Questions 1 - 10)

Directions: Write True or False for each statement and provide a rigorous mathematical justification based on circle theorems.

- Q1:** Two chords AB and CD of a circle are each at a distance of 4 cm from the center O . Then $AB = CD$.
- Q2:** Through three collinear points, a circle of unique radius can always be drawn.
- Q3:** A circle of radius 3 cm can be drawn through two distinct points A and B such that the segment length $AB = 6$ cm.
- Q4:** Two chords of a circle of lengths 10 cm and 8 cm are at distances 8.0 cm and 3.5 cm, respectively, from the center.
- Q5:** If AOB is a diameter of a circle and C is any point on the boundary of the circle, then $AC^2 + BC^2 = AB^2$.
- Q6:** The angles subtended by a chord at any two points lying on the boundary of a circle are always equal.
- Q7:** Two congruent circles with centers O and O' intersect at two distinct points A and B . Then the angle subtended at the centers are equal, i.e., $\angle AOB = \angle AO'B$.
- Q8:** If A, B, C, D are four points such that $\angle BAC = 30^\circ$ and $\angle BDC = 60^\circ$, then D must be the center of the circle passing through the points A, B , and C .
- Q9:** If A, B, C and D are four points such that $\angle BAC = 45^\circ$ and $\angle BDC = 45^\circ$, then the four points A, B, C, D are concyclic.
- Q10:** A cyclic quadrilateral $ABCD$ can be drawn such that the interior angle measures are $\angle A = 90^\circ$, $\angle B = 70^\circ$, $\angle C = 95^\circ$, and $\angle D = 105^\circ$.

Part B: Multiple Choice Questions (Questions 11 - 20)

Directions: Solve and select the correct option from the choices provided.

- Q11:** AD is a diameter of a circle and AB is a chord. If $AD = 34$ cm and $AB = 30$ cm, the perpendicular distance of the chord AB from the center of the circle is: (A) 17 cm (B) 15 cm (C) 4 cm (D) 8 cm
- Q12:** In a circle of radius 5 cm, a chord AB of length 8 cm is drawn. If $OD \perp AB$ intersects AB at C and the circle at D , then the length of the segment CD is: (A) 2 cm (B) 3 cm (C) 4 cm (D) 5 cm
- Q13:** If $AB = 12$ cm, $BC = 16$ cm, and $AB \perp BC$, then the radius of the unique circle passing through the three points A, B , and C is: (A) 6 cm (B) 8 cm (C) 10 cm (D) 12 cm
- Q14:** In a circle with center O , the angle subtended by an arc at the center is 150° , and OP and OQ are perpendiculars on two equal chords AB and CD . If $\angle POQ = 150^\circ$, then the measure of $\angle APQ$ is: (A) 30° (B) 75° (C) 15° (D) 60°
- Q15:** If a diameter AOB of a circle is drawn and a point C on the semicircle satisfies $AC = BC$, then the angle $\angle CAB$ is equal to: (A) 30° (B) 60° (C) 90° (D) 45°

- Q16:** In a circle, if the angle subtended at the center by chord AB is $\angle OAB = 40^\circ$ (where O is the center), then the angle subtended on the major segment, $\angle ACB$, is: (A) 50° (B) 40° (C) 60° (D) 70°
- Q17:** In a circle with center O , if the inscribed angles are such that $\angle DAB = 60^\circ$ and $\angle ABD = 50^\circ$, then the angle $\angle ACB$ is equal to: (A) 60° (B) 50° (C) 70° (D) 80°
- Q18:** $ABCD$ is a cyclic quadrilateral such that AB is a diameter of the circle circumscribing it. If $\angle ADC = 140^\circ$, then the measure of $\angle BAC$ is: (A) 80° (B) 50° (C) 40° (D) 30°
- Q19:** BC is a diameter of a circle with center O , and A is a point on the circle such that $\angle BAO = 60^\circ$. If D is a point on the remaining arc, then the value of $\angle ADC$ is: (A) 30° (B) 45° (C) 60° (D) 120°
- Q20:** If the central angle $\angle AOB = 90^\circ$ and the inscribed angle $\angle ABC = 30^\circ$ are given, then the measure of $\angle CAO$ is: (A) 30° (B) 45° (C) 90° (D) 60°

Part C: Calculation-Based Problems (Questions 21 - 32)

Directions: Show all calculation steps systematically. State the theorem or property applied in each justification.

- Q21:** A chord is at a distance of 5 cm away from the center of a circle of radius 13 cm. Find the exact length of the chord.
- Q22:** An arc of a circle subtends an angle of 70° at the center. What is the measure of the angle subtended by this same arc at any point on the remaining part of the circle?
- Q23:** The diameter of a circle is 26 cm. A chord of length 24 cm is drawn inside this circle. Find the perpendicular distance from the center of the circle to this chord.
- Q24:** A circle has a radius of 15 cm. A chord is drawn inside it at a perpendicular distance of 9 cm from the center of the circle. Calculate the length of the chord.
- Q25:** In a circle, a chord of length 16 cm is located at a perpendicular distance of 6 cm from the center. Find the radius of this circle.
- Q26:** A cyclic quadrilateral $ABCD$ is inscribed in a circle. If $\angle A$ measures 75° , find the measure of the opposite angle $\angle C$. If $\angle B$ measures 110° , find the measure of the opposite angle $\angle D$.
- Q27:** A quadrilateral $PQRS$ is inscribed in a circle. If its opposite angles are given as $\angle P = (2x + 10)^\circ$ and $\angle R = (3x - 20)^\circ$, find:
- (i) The value of the variable x .
 - (ii) The actual degree measures of $\angle P$ and $\angle R$.
- Q28:** Two parallel chords of lengths 10 cm and 24 cm lie on the same side of the center of a circle. If the perpendicular distance between the two chords is 7 cm, find the radius of the circle.
- Q29:** An isosceles triangle ABC with $AB = AC$ is inscribed in a circle of radius 5 cm. If the base $BC = 8$ cm, find the length of the equal sides AB and AC .
- Q30:** Find the value of the unknown angle x in a circle where a cyclic quadrilateral $ABCD$ has one diagonal as diameter AC , and the angle $\angle ADC = 100^\circ$.
- Q31:** In a circle with center O , the chords AB and AC subtend angles of 90° and 150° , respectively, at the center. Find the angle $\angle BAC$, if AB and AC lie on opposite sides of the center.
- Q32:** A cyclic quadrilateral has side lengths measuring 5, 5, 12, and 12 units in consecutive order. Calculate the total area of the quadrilateral.

Part D: Geometric Proofs (Questions 33 - 45)

Directions: Write formal geometric proofs using the perfect 5-step CBSE answer-writing structure. State all theorems used as reasons.

- Q33:** Prove that the perpendicular bisector of a chord of a circle passes directly through the center of the circle.
- Q34:** Prove that a rectangle is the only parallelogram that can be inscribed inside a circle.
- Q35:** If a rectangle is inscribed in a circle, prove that the point of intersection of its diagonals must lie exactly at the center of the circle.
- Q36:** Imagine drawing all possible chords of a fixed length in a circle. Prove that the midpoints of all these chords always form a concentric circle.

- Q37:** In a circle with center O , two chords AB and AC are congruent (equal in length). Prove that the center of the circle lies on the angle bisector of $\angle BAC$.
- Q38:** Prove that there is no chord of a circle that can be longer than its diameter.
- Q39:** Let A be any point lying within a given circle with center O . Prove that the shortest chord of the circle that passes through this point A is the one that is perpendicular to OA .
- Q40:** Prove that the angle subtended by an arc at the center of a circle is double the angle subtended by it at any point on the remaining part of the circle.
- Q41:** If two equal chords of a circle intersect, prove that the segments of one chord are separately equal to the corresponding segments of the other chord.
- Q42:** Prove that the sum of either pair of opposite angles of a cyclic quadrilateral is 180° .
- Q43:** If the non-parallel sides of a trapezium are equal, prove that the trapezium is cyclic.
- Q44:** Two circles with centers O and O' intersect at two distinct points A and B . A line PQ is drawn parallel to the line segment joining the centers OO' through the point A , intersecting the circles at P and Q . Prove that $PQ = 2OO'$.
- Q45:** In a circle, two chords CC' and DD' are drawn perpendicular to a diameter AB . Prove that the segment MM' joining the midpoints of the chords CD and $C'D'$ is perpendicular to the diameter AB .